

NFS (NETWORK FILE SYSTEM) SERVER

TASKS

1. Install the nfs-kernel-server package on your Linux VM and create a shared directory named /srv/music_share for sharing music files.

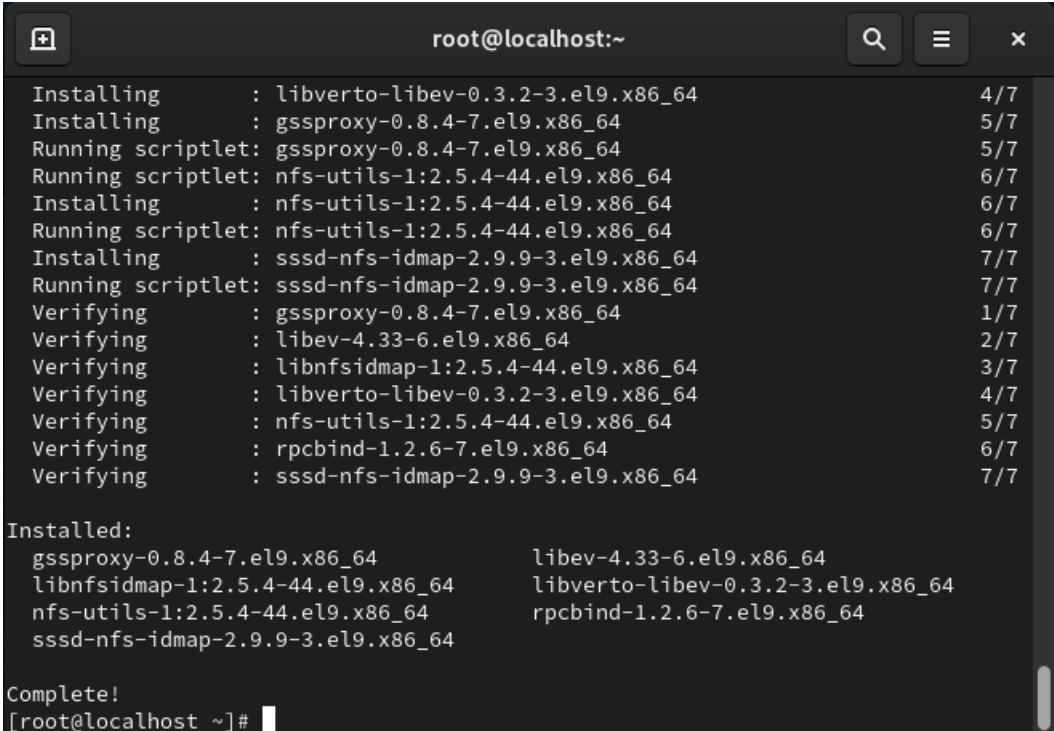
Install NFS Server and Create Shared Directory

Step 1: Become Root

su -

Step 2: Install NFS Server

dnf install nfs-utils -y



```
root@localhost:~  
Installing      : libverto-libev-0.3.2-3.el9.x86_64 4/7  
Installing      : gssproxy-0.8.4-7.el9.x86_64      5/7  
Running scriptlet: gssproxy-0.8.4-7.el9.x86_64      5/7  
Running scriptlet: nfs-utils-1:2.5.4-44.el9.x86_64  6/7  
Installing      : nfs-utils-1:2.5.4-44.el9.x86_64  6/7  
Running scriptlet: nfs-utils-1:2.5.4-44.el9.x86_64  6/7  
Installing      : sssd-nfs-idmap-2.9.9-3.el9.x86_64 7/7  
Running scriptlet: sssd-nfs-idmap-2.9.9-3.el9.x86_64 7/7  
Verifying       : gssproxy-0.8.4-7.el9.x86_64      1/7  
Verifying       : libev-4.33-6.el9.x86_64          2/7  
Verifying       : libnfsidmap-1:2.5.4-44.el9.x86_64 3/7  
Verifying       : libverto-libev-0.3.2-3.el9.x86_64 4/7  
Verifying       : nfs-utils-1:2.5.4-44.el9.x86_64  5/7  
Verifying       : rpcbind-1.2.6-7.el9.x86_64        6/7  
Verifying       : sssd-nfs-idmap-2.9.9-3.el9.x86_64 7/7  
  
Installed:  
gssproxy-0.8.4-7.el9.x86_64      libev-4.33-6.el9.x86_64  
libnfsidmap-1:2.5.4-44.el9.x86_64 libverto-libev-0.3.2-3.el9.x86_64  
nfs-utils-1:2.5.4-44.el9.x86_64  rpcbind-1.2.6-7.el9.x86_64  
sssd-nfs-idmap-2.9.9-3.el9.x86_64  
  
Complete!  
[root@localhost ~]#
```

**Step
3:**

Enable and Start Services

systemctl enable nfs-server rpcbind

systemctl start rpcbind

systemctl start nfs-server

Step 4: Check Status

systemctl status nfs-server

```
[root@localhost ~]# systemctl status nfs-server
● nfs-server.service - NFS server and services
   Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; enabled; prese>
   Active: active (exited) since Fri 2026-07-17 14:48:53 IST; 13s ago
     Docs: man:rpc.nfsd(8)
           man:exportfs(8)
   Process: 18673 ExecStartPre=/usr/sbin/exportfs -r (code=exited, status=0/SU>
   Process: 18674 ExecStart=/usr/sbin/rpc.nfsd (code=exited, status=0/SUCCESS)
   Process: 18690 ExecStart=/bin/sh -c if systemctl -q is-active gssproxy; the>
   Main PID: 18690 (code=exited, status=0/SUCCESS)
      CPU: 52ms

Jul 17 14:48:53 localhost.localdomain systemd[1]: Starting NFS server and servi>
Jul 17 14:48:53 localhost.localdomain systemd[1]: Finished NFS server and servi>
lines 1-13/13 (END)
```

**Step 5:
Create**

Shared Directory

```
mkdir -p /srv/music_share
```

Step 6: Give Permissions

```
chmod 777 /srv/music_share
```

Verify:

```
ls -ld /srv/music_share
```

```
[root@localhost ~]# ls -ld /srv/music_share
drwxrwxrwx. 2 root root 6 Jul 17 14:49 /srv/music_share
[root@localhost ~]#
```

2.Configure

/etc/exports to share /srv/music_share with read and write permissions to a specific client IP (use your own second VM or localhost for testing), then restart the NFS service.

Configure /etc/exports

Step 1: Edit Exports File

```
vi /etc/exports
```

Add this line (localhost testing):

If you have another VM:

```
/srv/music_share 192.168.231.128(rw,sync,no_root_squash)
```

Save:

Esc

:wq

Step 2: Export the Share

```
exportfs -rav
```

Step 3: Restart NFS

```
systemctl restart nfs-server
```

Step 4: Verify Export

```
exportfs -v
```

```
[root@localhost ~]# systemctl restart nfs-server
[root@localhost ~]# exportfs -v
/srv/music_share
198.168.231.128(sync,wdelay,hide,no_subtree_check,sec=sys,rw,secure,no_root_squash,no_all_squash)
[root@localhost ~]#
```

3. On a client

machine, mount the NFS share from your server to a local directory named `~/mnt/playlist` and verify you can create and access files in the shared folder.

Mount NFS Share

Step 1: Create Mount Point

```
mkdir -p /mnt/playlist
```

Step 2: Mount

For localhost:

```
mount -t nfs localhost:/srv/music_share /mnt/playlist
```

Step 3: Verify

```
mount | grep nfs
```

Step 4: Create File

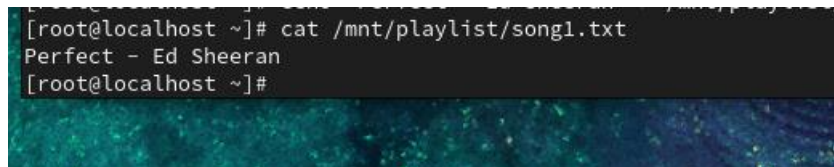
```
touch /mnt/playlist/song1.txt
```

Write content:

```
echo "Perfect - Ed Sheeran" > /mnt/playlist/song1.txt
```

Verify:

```
cat /mnt/playlist/song1.txt
```



```
[root@localhost ~]# cat /mnt/playlist/song1.txt
Perfect - Ed Sheeran
[root@localhost ~]#
```

4. Simulate a Zomato-style shared menu folder: add a subfolder called 'restaurant_menus' inside your NFS share, copy any 2 PDF or image files into it, and confirm they are visible from the client mount.

Create Restaurant Menu Folder

Step 1: Create Folder

```
mkdir /srv/music_share/restaurant_menus
```

Step 2: Copy Image or PDF

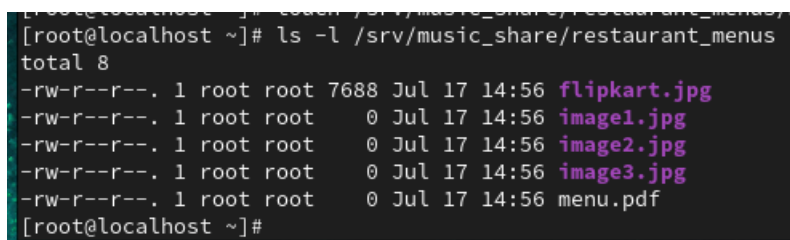
```
touch /srv/music_share/restaurant_menus/menu.pdf
```

Step 3: Verify

```
ls -l /srv/music_share/restaurant_menus
```

Step 4: Check From Client Mount

```
ls -l /mnt/playlist/restaurant_menus
```



```
[root@localhost ~]# touch /srv/music_share/restaurant_menus/menu.pdf
[root@localhost ~]# ls -l /srv/music_share/restaurant_menus
total 8
-rw-r--r--. 1 root root 7688 Jul 17 14:56 flipkart.jpg
-rw-r--r--. 1 root root    0 Jul 17 14:56 image1.jpg
-rw-r--r--. 1 root root    0 Jul 17 14:56 image2.jpg
-rw-r--r--. 1 root root    0 Jul 17 14:56 image3.jpg
-rw-r--r--. 1 root root    0 Jul 17 14:56 menu.pdf
[root@localhost ~]#
```

5. Un-mount the NFS share from the client and update your /etc/exports file to restrict access to read-only, then remount and test that you cannot create

or modify files in the shared folder.

Hint:
Use the 'ro' option in /etc/exports for read-only access.

Read-Only Share

Step 1: Unmount

```
umount /mnt/playlist
```

Step 2: Edit Exports

```
vi /etc/exports
```

Change:

```
/srv/music_share 127.0.0.1(rw,sync,no_root_squash)
```

to

```
/srv/music_share 127.0.0.1(ro,sync,no_root_squash)
```

Save:

Esc

:wq

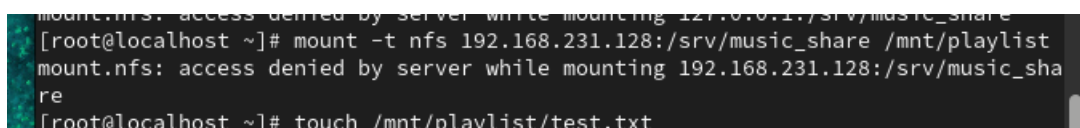
Step 3: Apply Changes

```
exportfs -rav
```

```
systemctl restart nfs-server
```

Step 4: Mount Again

```
mount -t nfs 127.0.0.1:/srv/music_share /mnt/playlist
```

A terminal window screenshot showing an NFS mount attempt. The first line shows an error: "mount.nfs: access denied by server while mounting 127.0.0.1:/srv/music_share". The second line shows the command: "[root@localhost ~]# mount -t nfs 192.168.231.128:/srv/music_share /mnt/playlist". The third line shows another error: "mount.nfs: access denied by server while mounting 192.168.231.128:/srv/music_share". The fourth line shows the command: "[root@localhost ~]# touch /mnt/playlist/test.txt".

```
mount.nfs: access denied by server while mounting 127.0.0.1:/srv/music_share
[root@localhost ~]# mount -t nfs 192.168.231.128:/srv/music_share /mnt/playlist
mount.nfs: access denied by server while mounting 192.168.231.128:/srv/music_share
[root@localhost ~]# touch /mnt/playlist/test.txt
```

Step

5: Test Write Access

```
touch /mnt/playlist/test.txt
```

```
[root@localhost ~]# touch /mnt/playlist/test.txt  
[root@localhost ~]#
```